

ABEL SANTAMARIA, Cuba

WMO: 783465

Lat: 22.492N

Lon: 79.944W

Elev: 103

StdP: 100.09

Time Zone: -5.00 (CUB)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	12.0	7.8	6.6	13.4	9.0	7.2	15.5	9.7	24.2	8.9	24.4	1.0	270	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.2	23.8	33.0	23.9	32.2	24.0	26.0	30.0	25.6	29.9	25.2	29.8	4.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.2	28.2	24.2	19.3	27.5	24.0	19.2	27.4	80.3	30.0	78.8	30.1	77.4	30.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.3	6.9	34.8	1.7	0.6	5.7	35.3	4.7	35.6	3.7	35.9	2.5	36.4		
(5)			6.4	27.3	1.9	1.1	5.1	28.1	4.0	28.7	2.9	29.3	1.6	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	21.2	21.7	22.3	24.5	25.6	26.8	27.4	27.5	26.9	25.7	22.8	22.3		
	DBStd	3.03	2.79	2.88	2.85	1.92	1.50	1.04	0.82	0.81	0.71	1.53	2.16	2.83		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	30	9	8	6	0	0	0	0	0	0	0	1	7		
	CDD10.0	5317	347	328	381	434	483	506	539	541	507	486	384	383		
	CDD18.3	2306	98	102	128	184	224	256	281	283	257	227	135	131		
	CDH23.3	18621	569	709	1023	1542	1869	2266	2678	2591	2142	1648	813	770		
(13)	CDH26.7	6949	118	187	322	577	734	932	1191	1149	873	542	154	170		
(14)	Wind	WSAvg	3.0	3.0	3.3	3.6	3.6	3.2	2.8	2.7	2.3	2.7	2.9	2.9		
(15) Precipitation	PrecAvg	1288	36	32	39	62	162	179	148	168	191	161	66	42		
	PrecMax	1630	76	117	84	118	328	362	216	249	348	480	180	203		
	PrecMin	840	8	7	14	21	60	83	86	109	111	52	8	9		
	PrecStd	175	18	24	16	21	75	65	37	32	48	87	45	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	31.2	32.2	33.2	34.0	33.8	34.2	34.1	33.2	32.2	30.9	31.0		
		MCWB	20.8	20.7	21.0	21.6	22.3	24.1	23.9	24.3	24.3	24.4	23.7	22.4		
	2%	DB	29.1	30.1	31.1	32.2	32.8	32.9	33.2	33.2	32.5	31.5	29.5	29.9		
		MCWB	21.0	20.7	20.6	21.5	22.5	24.2	24.2	24.4	24.5	24.4	23.2	22.5		
	5%	DB	28.0	29.0	30.0	31.2	31.2	32.1	32.8	32.9	32.0	30.9	28.2	28.8		
		MCWB	21.2	20.7	20.9	21.4	22.5	24.4	24.3	24.4	24.6	24.4	22.6	22.2		
	10%	DB	26.9	27.9	28.8	30.1	30.8	31.1	32.0	32.0	31.1	29.9	27.2	27.2		
		MCWB	20.9	20.8	20.5	21.5	22.5	24.3	24.4	24.6	24.7	24.3	21.9	21.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.2	23.2	24.0	25.0	26.1	26.1	26.4	26.2	26.0	25.0	24.3		
		MCDB	26.7	27.1	28.2	29.1	29.8	30.1	30.7	30.5	29.7	29.4	28.2	27.6		
	2%	WB	22.5	22.4	22.6	23.3	24.2	25.5	25.6	25.9	25.7	25.4	24.3	23.6		
		MCDB	26.1	26.3	27.4	28.3	29.0	29.9	30.3	30.4	29.6	29.1	27.3	27.1		
	5%	WB	22.0	21.9	21.9	22.7	23.7	25.0	25.2	25.5	25.3	25.0	23.7	23.1		
		MCDB	25.5	25.8	26.8	27.7	28.4	29.9	30.1	30.2	29.6	29.0	26.7	26.6		
	10%	WB	21.5	21.5	21.4	22.3	23.4	24.6	24.9	25.1	25.0	24.6	23.3	22.6		
		MCDB	25.0	25.5	26.2	27.2	28.0	29.4	29.8	30.0	29.6	28.3	26.1	25.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	12.0	11.9	10.6	9.5	10.2	10.1	9.5	8.3	9.3	9.9		
		MCDBR	11.6	12.5	13.0	12.9	11.9	10.4	10.9	10.9	10.4	9.7	9.9	10.6		
		MCWBR	5.1	5.2	4.8	4.4	4.2	3.9	3.9	4.0	4.1	4.0	4.8	4.7		
	5% WB	MCDBR	9.9	10.7	11.5	11.4	10.0	9.7	10.0	10.0	9.5	8.5	7.8	8.9		
		MCWBR	4.6	4.7	4.7	4.2	3.9	4.0	4.1	4.2	4.2	4.0	4.2	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.360	0.376	0.397	0.423	0.458	0.464	0.441	0.416	0.404	0.376	0.377		
	taud		2.528	2.516	2.460	2.395	2.368	2.301	2.272	2.355	2.434	2.454	2.486	2.475		
	Ebn at Noon		914	931	927	906	874	837	832	855	876	874	883	874		
	Edh at Noon		95	102	113	122	125	132	137	126	114	108	99	98		
(44) All-Sky Solar Radiation	RadAvg		3.95	4.83	5.64	6.19	5.98	5.69	6.22	5.97	5.35	4.58	4.07	3.69		
	RadStd		0.28	0.16	0.23	0.20	0.60	0.42	0.34	0.22	0.25	0.32	0.22	0.30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page